

1. Document Control

- System Name: Application

2. System Overview

- Application processes and transforms customer and order CSV data into Delta tables using Delta Live Tables.
- The system performs ingestion, cleansing, transformation, SCD Type 2 change tracking, and aggregation to enable analytical workloads supporting daily customer spend summaries.

3. User Roles

- Data Engineer: Configures and manages pipelines.
- Data Analyst: Consumes aggregate and transactional data.
- System Operator: Monitors data pipeline execution.
- No other roles specified in requirements.

4. Functional Scope

- Ingest customer and order source data from specified filesystem paths.
- Cleanse data to remove "Null"/Null literals and duplicates from both sources.
- Add computed TotalAmount field to the order table.
- Join customer and order for unified transactions view in ordersummary.
- Apply SCD Type 2 logic on customer data in ordersummary.
- Aggregate ordersummary to produce customeraggregatespend (sum of TotalAmount by Name and Date).
- All steps implemented in sequence via Delta Live Tables.

5. Functional Requirements

FR-APP-001 — Ingest Customer Data

- Description: Load customer CSV from /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata to Delta table 'customer'.
- Preconditions: Customer CSV present at given path.
- Main Flow:

1. Read customer CSV file from specified location.

2. Load data into Delta table named 'customer'.

- Alternate Flows: None.
- Acceptance Criteria: All customer records from the CSV are loaded into the Delta table without format or integrity errors.

FR-APP-002 — Ingest Order Data

- Description: Load order CSV from /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata to Delta table 'order'.
- Preconditions: Order CSV present at given path.
- Main Flow:

1. Read order CSV file from specified location.
2. Load data into Delta table named 'order'.

- Alternate Flows: None.
- Acceptance Criteria: All order records from the CSV are loaded into the Delta table without format or integrity errors.

FR-APP-003 — Data Cleansing for Customer

- Description: Cleanse customer table by removing "Null"/Null literals and duplicate records.
- Preconditions: Customer table loaded.
- Main Flow:

1. Scan all records in 'customer' table.
2. Remove any values that are literal "Null" or system nulls.
3. Remove duplicate entries based on all columns.

- Alternate Flows: None.
- Acceptance Criteria: Customer table contains only unique, valid records; all "Null"/Null values are removed.

FR-APP-004 — Data Cleansing for Order

- Description: Cleanse order table by removing "Null"/Null literals and duplicate records.
- Preconditions: Order table loaded.
- Main Flow:

1. Scan all records in 'order' table.
2. Remove any values that are literal "Null" or system nulls.
3. Remove duplicate entries based on all columns.

- Alternate Flows: None.
- Acceptance Criteria: Order table contains only unique, valid records; all "Null"/Null values are removed.

FR-APP-005 — Add Computed TotalAmount

- Description: Compute TotalAmount for each order as $\text{PricePerUnit} \times \text{Qty}$.
- Preconditions: Order table loaded and cleansed.
- Main Flow:
 1. For each order, calculate TotalAmount as $\text{PricePerUnit} \times \text{Qty}$.
 2. Add TotalAmount column to order table.
- Alternate Flows: None.
- Acceptance Criteria: All records in order table have a valid TotalAmount reflecting unit price $\times$ quantity.

FR-APP-006 — Create Ordersummary Table

- Description: Join customer and order tables to create ordersummary table.
- Preconditions: customer and order tables prepared.
- Main Flow:
 1. Join customer and order on CustId.
 2. Insert joined records into new table ordersummary.
- Alternate Flows: None.
- Acceptance Criteria: Ordersummary combines all order and customer attributes per transaction as specified.

FR-APP-007 — Implement SCD Type 2 for Customer in Ordersummary

- Description: Apply SCD Type 2 logic on customer attributes in ordersummary.
- Preconditions: Ordersummary table is being populated.
- Main Flow:
 1. Identify changed attributes for each customer in ordersummary.
 2. Mark previous version inactive, insert new as active.
 3. Populate StartDate/EndDate and Active status as per SCD2 conventions.
- Alternate Flows: If no change, retain previous status.
- Acceptance Criteria: Ordersummary retains full change history for customer data with appropriate versioning.

FR-APP-008 — Create Customer Aggregate Spend Table

- Description: Aggregate total spend per customer by Name and Date in customeraggregatespend table.
- Preconditions: Ordersummary table populated with TotalAmount.
- Main Flow:

1. Group ordersummary by Name and Date.

2. Calculate sum of TotalAmount for each group.

3. Write results to customeraggregatespend table.

- Alternate Flows: None.
- Acceptance Criteria: Each record in customeraggregatespend reflects accurate daily spend totals per customer.

FR-APP-009 — End-to-end Pipeline Execution in Delta Live Tables

- Description: Execute all loading, cleansing, calculation, joining, SCD handling, and aggregation using Delta Live Tables.
- Preconditions: Delta Live Tables environment available.
- Main Flow:

1. Configure Delta Live Tables pipeline for all steps above.

2. Trigger execution and monitor for errors.

- Alternate Flows: None.
- Acceptance Criteria: Pipeline completes all steps in sequence with no interruptions or data loss.

6. Business Rules

- TotalAmount must be accurately computed as $\text{PricePerUnit} \times \text{Qty}$.
- No record in customer/order tables may contain literal "Null" or be duplicated.
- Join between customer and order is always on CustId.
- SCD Type 2 must capture full history of changes for customer data.
- Aggregated spend must always be grouped by Name and Date.

7. Error Handling

- Fail record ingestion for badly formatted CSV rows, log error.
- Skip aggregation if upstream ordersummary data is incomplete.
- Capture and report any join/mapping mismatches (e.g., missing CustId).

8. Dependencies

- Pipeline depends on availability of both customer and order CSV files.

- Downstream aggregation depends on successful join and SCD tracking.
- All logic is tied to Delta Live Tables implementation.

9. Acceptance Criteria

- 100% ingestion of valid, unique customer and order data from respective CSVs.
- All "Null"/Null records and duplicates eliminated.
- TotalAmount present and accurate across orders and aggregates.
- Ordersummary correctly joins and reflects customer history with SCD2 changes.
- Customeraggregatespend table holds true sum per Name/Date group.
- Delta Live Tables pipeline completes end-to-end process without errors.

10. Architecture Diagram (Graphviz)

[Diagram render failed (graphviz): fileName=<_io.BytesIO object at 0x0000021CB1BB7330>
identity=[ImageReader@0x21cb24de490] cannot identify image file <_io.BytesIO object at
0x0000021CB1BB7330>]

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